Mini case, Lecture 3: Work or shirk?

Lecture 3, slides 41–42

Employee W/S (rows), manager M/N (columns); $w = 10$, $g = 2$, $v = 20$, $h = 1$. $p = \Pr(S)$, $q = \Pr(M)$. Numeric matrix (employee, manager): $(W, M) = (8, 9)$, $(W, N) = (8, 10)$, $(S, M) = (0, -1)$, $(S, N) = (10, -10)$.

(a) No pure-strategy NE. Every cell has a profitable unilateral deviation, and the deviations form a cycle:

$$(W, M) \xrightarrow{\text{Mg:}+h=1} (W, N) \xrightarrow{\text{E:}+g=2} (S, N) \xrightarrow{\text{Mg:}+(w-h)=9} (S, M) \xrightarrow{\text{E:}+(w-g)=8} (W, M).$$

Best responses: employee W vs M ($8 > 0$), S vs N ($10 > 8$); manager N vs W ($10 > 9$), M vs S ($-1 > -10$). No cell is a mutual best response. In general this holds whenever $0 < g < w$ and $0 < h < w$.

(b) Mixed NE. Both must mix (if one played pure, the other's unique best response is pure, contradicting (a)), so by the principle of indifference each is indifferent between his two actions.

$$\begin{aligned} \text{Employee:}\quad & u_E(W, q) = 8, \quad u_E(S, q) = 10(1 - q) && \Rightarrow\ 8 = 10 - 10q \ \Rightarrow\ q^* = 0.2, \\ \text{Manager:}\quad & u_{Mg}(M, p) = 9(1 - p) - p = 9 - 10p, \quad u_{Mg}(N, p) = 10 - 20p && \Rightarrow\ 10p = 1 \ \Rightarrow\ p^* = 0.1. \end{aligned}$$

The employee shirks with probability 0.1 and the manager monitors with probability 0.2. Payoffs: employee 8, manager $9 - 10(0.1) = 8$. This is the unique NE (the best responses cross once, at $(p, q) = (0.1, 0.2)$).

(c) $h = 0.5$**.** The employee's condition has no h: $q^* = 0.2$, unchanged. Manager: $9.5 - 10p = 10 - 20p \Rightarrow p^* = 0.05$. Shirking halves; monitoring does not change; the firm's payoff rises from 8 to 9. *Intuition:* each player's mix is pinned by the *opponent's* indifference. Monitoring must keep the employee indifferent, which depends only on g and w. Cheaper monitoring tempts the manager to monitor more, so employees shirk less until the manager is indifferent again, at $p = h/w$.

(d) $w = 20$**.** Employee: $18 = 20(1 - q) \Rightarrow q^* = 0.1$. Manager: $u_{Mg}(M, p) = -1$, $u_{Mg}(N, p) = -20p \Rightarrow p^* = 0.05$. Both halve: a higher wage raises what a caught shirker loses (less monitoring needed) and what an undetected shirker costs the firm (less shirking needed to justify monitoring). Payoffs: employee 18, firm -1. The firm buys the same fall in shirking as in (c) for 9 more.

(e) General formulas and policy. For $0 < g < w$, $0 < h < w$:

$$p^* = \frac{h}{w}, \qquad q^* = \frac{g}{w}, \qquad u_E^* = w - g, \qquad \pi(w, h) = v - w - \frac{hv}{w}.$$

Shirking depends only on the manager's trade-off (h vs w), monitoring only on the employee's (g vs w); v enters neither. $\partial\pi/\partial h = -v/w = -2$ and $\partial\pi/\partial w = -1 + hv/w^2 = -0.8 < 0$: a raise costs 1 for sure and saves only 0.2. Within the model the profit-maximizing wage is $\sqrt{hv} = \sqrt{20} \approx 4.47$ ($> g$), leaving the employee ≈ 2.47; the model ignores the participation constraint, which in practice sets the wage floor.

Recommendation. Cheaper or better monitoring, not a higher wage: halving h takes the firm from 8 to 9, while $w = 20$ takes it to -1. With detection probability δ, $\pi = v - w - hv/(\delta w)$, so what matters is the cost per detection h/δ. Better still, *commit* (Lecture 3, Stackelberg logic): announce and hard-wire random audits at rate $q = g/w = 0.2$; the employee then works ($p = 0$) and $\pi = v - w - hg/w = 9.8 > 8$. The commitment must be credible: ex post, once the employee works, monitoring costs h and finds nothing, so a manager who can revise the promise will not monitor (time inconsistency).

Assumptions to question in the Itaú case. (i) Perfect detection: software measures activity, not output ($\delta < 1$ raises p^* and q^*; mouse jigglers). (ii) Binary effort: real effort is continuous and multidimensional. (iii) The penalty is only the wage: dismissal also destroys future rents R, giving $q^* = g/(w + R)$; this is the repeated game behind efficiency wages (Lecture 6). (iv) No false positives and no legal limits: mistaken flags lower the return to working, and the LGPD and labour courts raise h and cap monitoring. (v) Identical employees: heterogeneous g calls for targeted, not random, audits. (vi) One-shot simultaneous timing: the relationship is repeated, employees know software is installed, and continuous automated monitoring has near-zero marginal h.